

Audit Report: Second staged 5G incident — healthcare private 5G slice

Item	Detail
Run ID	scenario2_20261010T050639Z
Scenario	scenario2 (DOCX §2.5)
Run started	2026-10-10T05:06:39.915152+00:00
Last entry recorded	2026-10-10T05:07:47.808011+00:00
Run status	 Incomplete — 1 of 4 stages recorded, run did not finish
Audit-trail integrity	 Intact
Source file	scenario2_20261010T050639Z.jsonl

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1. Executive summary

Key figures

Measure	Value
Stages recorded	1 of 4
Specialist agents used	1 (Technical)
Passages retrieved	14
Claims checked by the Verifier	6
Verifier outcomes	🟡 INCOMPLETE 4, 🟥 UNSUPPORTED 2
Evidence-backed conclusions (final stage)	0
Agent selection matching DOCX table	1 of 1

What happened

So far one stage has been released: at T0 intermittent latency and session drops in the private 5G slice (Technical joined). The Orchestrator chose the specialist agents from what each stage revealed, matching the official stage table at 1 of 1 stages. No security, critical-service or personal-data indicator has been raised yet. The agents quoted provisions mainly from 3GPP TS 23.501. The Verifier checked every claim against the separate Canonical KB: 4 INCOMPLETE and 2 UNSUPPORTED.

What follows

On the evidence retrieved, the incident is a network performance problem; no security indicator has been released. No notification duty with a time limit was found in the retrieved Indian provisions at this stage. No conclusion is yet fully evidence-backed: quotes match the authoritative text, but the in-force and amendment status of the sources has not been confirmed. Still open: core network telemetry (AMF/SMF logs) not yet released. This is decision support: legal interpretation and any regulatory action remain with qualified people.

The two paragraphs above are produced by the project's `src/audit/narrative.py` from the recorded entries; each sentence restates a recorded fact, quoted provision or Verifier outcome.

Observations for judges

1. **The run is incomplete.** Only 1 of 4 planned stages was recorded and there is no `run_completed` entry, so the run stopped before the remaining stages were processed. Everything below describes the stage(s) that were recorded.
2. **The audit trail is intact.** Every entry's SHA-256 hash and its link to the previous entry check out.
3. **Agent selection matched the official DOCX stage table at 1 of 1 stage(s).**
4. **No claim reached VERIFIED** (🟡 **INCOMPLETE** 4, 🟥 **UNSUPPORTED** 2). The quoted passages match the authoritative text, but the in-force and amendment status of the sources was not confirmed, so the system reports them as INCOMPLETE rather than overstating them; claims without a supporting passage are UNSUPPORTED and treated as preliminary.
5. **LLM claims: 0 accepted, 2 rejected** by the citation check before reaching the Verifier; only passages quoted from the knowledge base were verified. In 2 of the rejected claims the LLM did cite passages it had been offered, but wrote the IDs inside square brackets (e.g.

[3gpp_ts_23501:clause-5-45-2:1]), so they did not match the offered IDs exactly and were rejected as "cites no retrieved passage id".

6. **The run header records two different statements about generation:** `generator_model` is "none — agents quote retrieved passages; no generative model is used", while `llm` records `ollama · qwen2.5:7b-instruct`, which the agents used for reasoning. Both are reproduced as recorded.
 7. **Sources not ingested into the knowledge base:** `nciipc_rules_2013`, `international_policy_examples`.
 8. **The run was made from commit `6bcc7f30647919e4a1bd47491c85ea6236f95dc2` with uncommitted changes.**
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2. Run status and integrity

Entries in the audit trail

Seq	Entry	Recorded at
1	run_started	2026-10-10T05:06:39.915152+00:00
2	stage T0	2026-10-10T05:07:47.808011+00:00

Run completion

 No `run_completed` entry was recorded. The header planned 4 stages; 1 was recorded. The remaining stages were not run or not recorded, so this report covers only the recorded stage(s).

Integrity check

 **Intact.** Each entry stores the SHA-256 of its own content (`hash`) and the hash of the entry before it (`prev_hash`). All hashes were recomputed with `src.audit.trail.verify_chain` and match, so no entry was edited, removed or reordered. The full chain is in [Appendix E](#).

3. Run configuration

Run header

Field	Value
Seq	1
Run id	scenario2_20261010T050639Z
Type	run_started
Schema version	1.0
Recorded at	2026-10-10T05:06:39.915152+00:00
Scenario id	scenario2
Scenario title	Second staged 5G incident — healthcare private 5G slice
Docx section	§2.5
Stage count	4
Knowledge base note	null
Prev hash	00
Hash	c060cb26f5eaf83b10550e64a3b857a130b335fd7d75a22c1f7bf760aa0e8e22

Knowledge bases

Knowledge base	Live
canonical	✔ true
technical	✔ true
policy_legal	✔ true
cybersecurity	✔ true
privacy	✔ true
critical_infrastructure	✔ true
standards	✔ true
policy_gap	✔ true

Knowledge-base build

Field	Value
Ingestion manifest sha256	8dacfff6b9e8896e42a68f76b2e6e04cb240f2394378fe7ef5f45cfbc292e1cb
Built at	2026-10-09T20:28:46.292329+00:00
Embedding model	BAAI/bge-small-en-v1.5
Generator model	none — agents quote retrieved passages; no generative model is used
Sandbox	not executed — the ITU AI for Good Sandbox is not available in this environment

Not ingested:

- nciipc_rules_2013
- international_policy_examples

Language model

Field	Value
Provider	ollama
Model	qwen2.5:7b-instruct

Code version

Field	Value
Commit	6bcc7f30647919e4a1bd47491c85ea6236f95dc2
Uncommitted changes	true

4. Incident timeline

Stage	Scenario time	Information released	Agents active	Newly activated	Verifier outcomes	Flags raised
T0	2024-10-04T09:00:00Z	Intermittent latency and session drops in the private 5G slice.	Technical	Technical	🟡 INCOMPLETE 4, 🟥 UNSUPPORTED 2	none

5. Stage-by-stage analysis

5.1 Stage T0

Scenario time: 2024-10-04T09:00:00Z

Chunk: scenario2_T0

What was released

Intermittent latency and session drops in the private 5G slice.

New facts:

- A private 5G network slice is experiencing intermittent latency spikes.
- Session drops are occurring on the affected slice.
- The degradation is intermittent rather than a complete outage.
- No additional context about the cause or the service supported by the slice is available at this stage.

Information released only to specific agents:

- Technical (technical) only:
 - NOC dashboard: one-way latency on the private slice peaks at 180 ms against a 20 ms target; about 3% of PDU sessions on the slice were released abnormally in the last hour.

Prior chunks considered: none (first stage)

Official DOCX stage table	Text
Information released	Intermittent latency and session drops in the private 5G slice.
Expected change	Classify radio/core/slicing/service-quality symptoms; identify missing telemetry.

Agents selected by the Orchestrator

Item	Agents
Active	Technical (technical)
Newly activated	Technical (technical)
No longer active (conclusions carried forward)	none
Primary agents in the DOCX table	Technical (technical)
Matches the DOCX table	 Yes

Findings of the specialist agents

5.1.1 Technical agent

Decision summary: Technical assessment: Network performance degradation / potential service-quality failure. Affected components: Network Slice (performance layer), User Plane Function (UPF), Private 5G Network (dedicated slice). Further telemetry required to confirm root cause.

Mandate output:

- **Incident class:** Network performance degradation / potential service-quality failure

- **Affected components:**
- Network Slice (performance layer)
- User Plane Function (UPF)
- Private 5G Network (dedicated slice)

Search queries used:

1. Intermittent latency and session drops in the private 5G slice.
2. network slice quality of service, latency and packet delay monitoring
3. PDU session release and loss of user plane connectivity

Evidence retrieved (14 passages; full text in [Appendix A](#)):

#	Source	Section	Section title	Page	Relevance	Chunk ID
1	3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19	Clause 5.37.7.1	General	(empty)	0.7032	3gpp_ts_23501: clause-5-37-7-1:1
2	3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19	Clause 5.5.2	Connection Management	(empty)	0.7586	3gpp_ts_23501: clause-5-5-2:6
3	3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19	Clause 5.33.1	General	(empty)	0.7002	3gpp_ts_23501: clause-5-33-1:1
4	3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19	Clause 5.45.2	Packet delay monitoring	(empty)	0.7214	3gpp_ts_23501: clause-5-45-2:1
5	3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19	Clause 5.7.5.1	General	(empty)	0.7347	3gpp_ts_23501: clause-5-7-5-1:2
6	3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19	Clause 5.7.3.4	Packet Delay Budget	(empty)	0.6901	3gpp_ts_23501: clause-5-7-3-4:5
7	3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19	Clause 3.1	Definitions	(empty)	0.7158	3gpp_ts_23501: clause-3-1:14
8	3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19	Clause 5.17.3	Interworking with EPC in presence of Non-3GPP PDU Sessions	(empty)	0.7346	3gpp_ts_23501: clause-5-17-3:1
9	3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19	Clause 4.2.12	Architecture for Proximity based Services (ProSe) in 5GS	(empty)	0.682	3gpp_ts_23501: clause-4-2-12:1
10	3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19	Clause 5.33.3.3	GTP-U Path Measurement	(empty)	0.7141	3gpp_ts_23501: clause-5-33-3-3:4
11	3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19	Clause 5.6.4.3	Usage of IPv6 multi-homing for a PDU Session	(empty)	0.7141	3gpp_ts_23501: clause-5-6-4-3:1
12	3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19	Clause 5.27.1.12	Support for network timing synchronization status monitoring	(empty)	0.6762	3gpp_ts_23501: clause-5-27-1-12:1

#	Source	Section	Section title	Page	Relevance	Chunk ID
13	3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19	Clause 5.15.1	General	(empty)	0.7107	3gpp_ts_23501:clause-5-15-1:5
14	3GPP TS 33.501 V19.7.0 (2026-06) Security architecture and procedures for 5G system, Release 19	Clause M.4	Protection of management traffic between IAB-node and OAM	(empty)	0.7135	3gpp_ts_33501:clause-m-4:1

Claims and verification (6: ● INCOMPLETE 4, ● UNSUPPORTED 2; citations in full in Appendix B):

#	Claim	Verifier outcome	Verifier rationale
1	The incident involves Network Slice (performance layer), User Plane Function (UPF), Private 5G Network (dedicated slice).	● UNSUPPORTED	Checked against any retrieved Canonical KB passage: the authoritative text does not support the claim (the claim's content is not found in the passage).
2	Incident classification: Network performance degradation / potential service-quality failure.	● UNSUPPORTED	Checked against any retrieved Canonical KB passage: the authoritative text does not support the claim (the claim's content is not found in the passage).
3	[REFERENCE ONLY — not Indian law] 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.37.7.1 states: "The 5GS Packet Delay Variation is the variation of packet delay measured between UE and PSA UPF. The AF may send the requirement for Packet Delay Variation monitoring to 5GS together with the requirement for packet delay measurement, as described in clause 6.1.3.26 of TS 23.503 [45]. Upon AF request for Packet Delay Variation monitoring together with packet delay monitoring, the PCF triggers the ..."	● INCOMPLETE	Supported by the authoritative text (3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.37.7.1), but in-force status not verified and amendments not checked for this source.
4	[REFERENCE ONLY — not Indian law] 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.5.2 states: "for the non-3GPP access, these PDU Sessions are not released to enable the UE to move the PDU Sessions over the 3GPP access based on UE policies. The core network maintains the PDU Sessions but deactivates the N3 user plane connection for such PDU Sessions."	● INCOMPLETE	Supported by the authoritative text (3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.5.2; 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.5.2), but in-force status not verified and amendments not checked for this source.
5	[REFERENCE ONLY — not Indian law] 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.33.1 states: "The following features described in 5.33 may be used to enhance 5GS to support Ultra Reliable Low Latency Communication (URLLC): - Redundant transmission for high reliability communication. In this Release, URLLC applies to 3GPP access only. When a PDU Session is to serve URLLC QoS Flow, the UE and SMF should establish the PDU Session as always-on PDU Session as described in clause 5.6.13. NOTE ..."	● INCOMPLETE	Supported by the authoritative text (3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.33.1), but in-force status not verified and amendments not checked for this source.

#	Claim	Veri fier out co me	Verifier rationale
6	[REFERENCE ONLY — not Indian law] 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.45.2 states: "QoS Monitoring for packet delay allows for the measurement of UL packet delay, DL packet delay or round trip packet delay between UE and PSA UPF. The details of the QoS Monitoring for packet delay are described in clause 5.33.3. The PCF may calculate Packet Delay Variation (clause 5.37.7) and the round trip packet delay when UL and DL are on different QoS flows (clause 5.37.4) based on packet ..."	● INC OM PLE TE	Supported by the authoritative text (3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.45.2), but in-force status not verified and amendments not checked for this source.

Uncertainty noted by the agent:

- [LLM-assisted — ollama:qwen2.5:7b-instruct] The exact cause of the latency spikes and session drops is uncertain without further diagnostic information.

Missing facts noted by the agent:

- Core network telemetry (AMF/SMF logs) not yet released.
- Radio access network (RAN) performance counters not yet available.
- Whether the performance degradation is isolated to one slice or system-wide is unknown.
- [LLM-assisted — ollama:qwen2.5:7b-instruct] The specific service supported by the private 5G slice and the potential impact on the service are not provided.

LLM-assisted reasoning (prompts and raw output in [Appendix C](#)):

Field	Value
Provider / model	ollama · qwen2.5:7b-instruct
Outcome	0 claim(s) accepted, 2 rejected; accepted claims go to the Verifier
Latency (ms)	63025.2
Tokens	input_tokens 2433, output_tokens 316

Reasoning summary:

The intermittent latency and session drops in the private 5G slice suggest issues with packet delay and potential congestion. The NOC dashboard indicates that the one-way latency is significantly higher than the target, and 3% of PDU sessions were abnormally released, which could be due to congestion or other issues.

Accepted claims:

None.

Rejected claims:

#	Claim proposed by the LLM	Passages it cited	Reason rejected
1	The high one-way latency (180 ms) and packet drops indicate potential congestion issues in the 5G network, which could be affecting the private slice's performance.	[3gpp_ts_23501:clause-5-45-2:1], [3gpp_ts_23501:clause-5-7-3-4:5]	cites no retrieved passage id
2	The 3% abnormal release of PDU sessions could be due to congestion or other issues, as non-GBR QoS flows are expected to experience packet drops in uncongested scenarios.	[3gpp_ts_23501:clause-5-7-3-4:5]	cites no retrieved passage id

Verification

Outcome counts:  INCOMPLETE 4,  UNSUPPORTED 2

Cross-domain links:

None.

Cross-domain link details:

None.

Conflicts:

None.

Conflict details:

None.

Missing evidence:

- [technical] Core network telemetry (AMF/SMF logs) not yet released.
- [technical] Radio access network (RAN) performance counters not yet available.
- [technical] Whether the performance degradation is isolated to one slice or system-wide is unknown.
- [technical] [LLM-assisted — ollama:qwen2.5:7b-instruct] The specific service supported by the private 5G slice and the potential impact on the service are not provided.

Coordinator assessment

Chunk: scenario2_T0 · Active agents: Technical (technical)

Incident flags:

Flag	Raised
Cyber event suspected (cyber_event_suspected)	false
Critical infrastructure flagged (cii_flagged)	false
Personal-data exposure suspected (data_exposure_suspected)	false

What changed from the previous stage (1):

1. Initial assessment — no prior chunk to compare against.

Confirmed facts (5):

1. [Incident fact — scenario2_T0] Intermittent latency and session drops in the private 5G slice.

2. [New fact — scenario2_T0] A private 5G network slice is experiencing intermittent latency spikes.
3. [New fact — scenario2_T0] Session drops are occurring on the affected slice.
4. [New fact — scenario2_T0] The degradation is intermittent rather than a complete outage.
5. [New fact — scenario2_T0] No additional context about the cause or the service supported by the slice is available at this stage.

Evidence-backed conclusions (0):

None.

Uncertain or preliminary conclusions (6):

1. [TECHNICAL | UNSUPPORTED — insufficient evidence; preliminary only] The incident involves Network Slice (performance layer), User Plane Function (UPF), Private 5G Network (dedicated slice). (Verification note: Checked against any retrieved Canonical KB passage: the authoritative text does not support the claim (the claim's content is not found in the passage).)
2. [TECHNICAL | UNSUPPORTED — insufficient evidence; preliminary only] Incident classification: Network performance degradation / potential service-quality failure. (Verification note: Checked against any retrieved Canonical KB passage: the authoritative text does not support the claim (the claim's content is not found in the passage).)
3. [TECHNICAL | INCOMPLETE] [REFERENCE ONLY — not Indian law] 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.37.7.1 states: "The 5GS Packet Delay Variation is the variation of packet delay measured between UE and PSA UPF. The AF may send the requirement for Packet Delay Variation monitoring to 5GS together with the requirement for packet delay measurement, as described in clause 6.1.3.26 of TS 23.503 [45]. Upon AF request for Packet Delay Variation monitoring together with packet delay monitoring, the PCF triggers the ..." (Verification note: Supported by the authoritative text (3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.37.7.1), but in-force st...)
4. [TECHNICAL | INCOMPLETE] [REFERENCE ONLY — not Indian law] 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.5.2 states: "for the non-3GPP access, these PDU Sessions are not released to enable the UE to move the PDU Sessions over the 3GPP access based on UE policies. The core network maintains the PDU Sessions but deactivates the N3 user plane connection for such PDU Sessions." (Verification note: Supported by the authoritative text (3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.5.2; 3GPP TS 23.501 V19....)
5. [TECHNICAL | INCOMPLETE] [REFERENCE ONLY — not Indian law] 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.33.1 states: "The following features described in 5.33 may be used to enhance 5GS to support Ultra Reliable Low Latency Communication (URLLC): - Redundant transmission for high reliability communication. In this Release, URLLC applies to 3GPP access only. When a PDU Session is to serve URLLC QoS Flow, the UE and SMF should establish the PDU Session as always-on PDU Session as described in clause 5.6.13. NOTE ..." (Verification note: Supported by the authoritative text (3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.33.1), but in-force stat...)
6. [TECHNICAL | INCOMPLETE] [REFERENCE ONLY — not Indian law] 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.45.2 states: "QoS Monitoring for packet delay allows for the measurement of UL packet delay, DL packet delay or round trip packet delay between UE and PSA UPF. The details of the QoS Monitoring for packet delay are described in clause 5.33.3. The PCF may calculate Packet Delay Variation

(clause 5.37.7) and the round trip packet delay when UL and DL are on different QoS flows (clause 5.37.4) based on packet ..." (Verification note: Supported by the authoritative text (3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.45.2), but in-force stat...)

Conflicting findings (0):

None.

Potential policy gaps (for expert review) (0):

None.

Cross-domain relationships (0):

None.

Relevant institutions (0):

None.

Open questions (5):

- 1. No conclusion is evidence-backed at this stage: no claim was found supported by an authoritative source whose in-force status and amendments have been checked. Every conclusion below is uncertain or preliminary.
- 2. [technical] Core network telemetry (AMF/SMF logs) not yet released.
- 3. [technical] Radio access network (RAN) performance counters not yet available.
- 4. [technical] Whether the performance degradation is isolated to one slice or system-wide is unknown.
- 5. [technical] [LLM-assisted — ollama:qwen2.5:7b-instruct] The specific service supported by the private 5G slice and the potential impact on the service are not provided.

Claim register (6):

#	A g e n t	Ou tco me	Claim	Rationale	Citations	C h u n k	Ca rri ed fo rwa rd	In c o n fli ct
1	Te c h ni c al	 UN SU PP OR TE D	The incident involves Network Slice (performance layer), User Plane Function (UPF), Private 5G Network (dedicated slice).	Checked against any retrieved Canonical KB passage: the authoritative text does not support the claim (the claim's content is not found in the passage).	—	sc e n ar io 2_ T 0	fal se	fa lse

#	Argument	Outcome	Claim	Rationale	Citations	Checked	Carried forward	In conflict
2	Technical	❌ UNSUPPORTED	Incident classification: Network performance degradation / potential service-quality failure.	Checked against any retrieved Canonical KB passage: the authoritative text does not support the claim (the claim's content is not found in the passage).	—	section 2_T0	false	false
3	Technical	🟡 INCOMPLET	[REFERENCE ONLY — not Indian law] 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.37.7.1 states: "The 5GS Packet Delay Variation is the variation of packet delay measured between UE and PSA UPF. The AF may send the requirement for Packet Delay Variation monitoring to 5GS together with the requirement for packet delay measurement, as described in clause 6.1.3.26 of TS 23.503 [45]. Upon AF request for Packet Delay Variation monitoring together with packet delay monitoring, the PCF triggers the ..."	Supported by the authoritative text (3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.37.7.1), but in-force status not verified and amendments not checked for this source.	3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.37.7.1	section 2_T0	false	false
4	Technical	🟡 INCOMPLET	[REFERENCE ONLY — not Indian law] 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.5.2 states: "for the non-3GPP access, these PDU Sessions are not released to enable the UE to move the PDU Sessions over the 3GPP access based on UE policies. The core network maintains the PDU Sessions but deactivates the N3 user plane connection for such PDU Sessions."	Supported by the authoritative text (3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.5.2; 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.5.2), but in-force status not verified and amendments not checked for this source.	3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.5.2	section 2_T0	false	false
5	Technical	🟡 INCOMPLET	[REFERENCE ONLY — not Indian law] 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.33.1 states: "The following features described in 5.33 may be used to enhance 5GS to support Ultra Reliable Low Latency Communication (URLLC): - Redundant transmission for high reliability communication. In this Release, URLLC applies to 3GPP access only. When a PDU Session is to serve URLLC QoS Flow, the UE and SMF should establish the PDU Session as always-on PDU Session as described in clause 5.6.13. NOTE ..."	Supported by the authoritative text (3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.33.1), but in-force status not verified and amendments not checked for this source.	3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.33.1	section 2_T0	false	false

#	A g e n t	Ou tco me	Claim	Rationale	Citations	C h u n k	Ca rri ed fo rwa rd	In c o n fli ct
6	Te c h ni c al	● INC OM PL ET E	[REFERENCE ONLY — not Indian law] 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.45.2 states: "QoS Monitoring for packet delay allows for the measurement of UL packet delay, DL packet delay or round trip packet delay between UE and PSA UPF. The details of the QoS Monitoring for packet delay are described in clause 5.33.3. The PCF may calculate Packet Delay Variation (clause 5.37.7) and the round trip packet delay when UL and DL are on different QoS flows (clause 5.37.4) based on packet ..."	Supported by the authoritative text (3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.45.2), but in-force status not verified and amendments not checked for this source.	3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.45.2	sc e n ar io 2_ T 0	fal se	fa lse

Link register:

None.

Human review required: Legal interpretation, regulatory decisions, and institutional action require qualified human judgment. This assessment is decision support only and does not replace legal counsel, regulators, or government decision-makers.

6. Notifications that may be due

Read from the quoted Indian provisions; each must be confirmed by a qualified person before it is sent.

No reporting duty with a recipient or time limit appears in the provisions quoted in the recorded stage(s).

7. Glossary

Verifier outcomes

Outcome	Meaning
 VERIFIED	Supported by the authoritative text, with in-force status and amendments checked.
 INCOMPLETE	Supported by the authoritative text, but in-force status, amendments or a related provision were not confirmed.
 UNSUPPORTED	The Canonical KB passages checked do not support the claim; preliminary only.
 CONFLICT	Authoritative sources disagree; the Coordinator records both sides.

Terms

Term	Meaning
Stage (T0, T1, ...)	One release of incident information; the scenario is revealed stage by stage.
Orchestrator	Chooses which specialist agents act at each stage from what that stage reveals.
Specialist agent	Retrieves passages from its own knowledge base and makes claims quoting them.
Verifier	Checks every claim against the separate Canonical knowledge base.
Coordinator	Combines verified findings into facts, conclusions, conflicts, gaps and open questions.
Chunk ID	Identifier of the stage, or of a passage in a knowledge base (document:section:part).
REFERENCE ONLY — not Indian law	International standard used as a reference point, never as an Indian obligation.
Hash chain	Each entry carries a SHA-256 of its content and of the previous entry, so edits are detectable.

Appendix A. Retrieved passages in full

Stage T0: Technical agent (14 passages)

A.T0.technical.1 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.37.7.1

Field	Value
Source title	3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19
Authority	3rd Generation Partnership Project (3GPP)
Jurisdiction	International
Document type	Standard
Section	Clause 5.37.7.1
Section title	General
Page	(empty)
Date issued	2026-09
Effective	null
Effective status	(empty)
Amendment checked	false
Amendment note	(empty)
Url	https://www.3gpp.org/ftp/Specs/archive/23_series/23.501/23501-j90.zip
Provenance note	Official archive 23501-j90.zip from 3gpp.org (latest Release 19 version on 2026-10-05; the DOCX does not fix a version). Body paragraphs only; tables and figures not ingested. Reference point only, not Indian law.
Chunk id	3gpp_ts_23501:clause-5-37-7-1:1
Relevance score	0.7032
Is stub	false

Excerpt:

The 5GS Packet Delay Variation is the variation of packet delay measured between UE and PSA UPF. The AF may send the requirement for Packet Delay Variation monitoring to 5GS together with the requirement for packet delay measurement, as described in clause 6.1.3.26 of TS 23.503 [45]. Upon AF request for Packet Delay Variation monitoring together with packet delay monitoring, the PCF triggers the QoS monitoring procedure and obtains the UL, DL or RT QoS Monitoring result from the SMF. After receiving the QoS Monitoring result, the PCF derives the 5GS Packet Delay Variation based on the QoS Monitoring result and then reports to the AF/NEF both packet delay measurements and Packet Delay Variation. The details of the QoS Monitoring and QoS Monitoring for packet delay are described in clauses 5.45 and 5.33.3. NOTE: The derivation of 5GS Packet Delay Variation by PCF, based on QoS Monitoring, is implementation dependent. The Packet Delay Variation calculation method needs to be the same within the

PLMN, based on operator policy. QoS Monitoring is used to obtain measurement of QoS parameters of individual QoS Flows.

A.T0.technical.2 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.5.2

Field	Value
Source title	3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19
Authority	3rd Generation Partnership Project (3GPP)
Jurisdiction	International
Document type	Standard
Section	Clause 5.5.2
Section title	Connection Management
Page	<i>(empty)</i>
Date issued	2026-09
Effective	null
Effective status	<i>(empty)</i>
Amendment checked	false
Amendment note	<i>(empty)</i>
Url	https://www.3gpp.org/ftp/Specs/archive/23_series/23.501/23501-j90.zip
Provenance note	Official archive 23501-j90.zip from 3gpp.org (latest Release 19 version on 2026-10-05; the DOCX does not fix a version). Body paragraphs only; tables and figures not ingested. Reference point only, not Indian law.
Chunk id	3gpp_ts_23501:clause-5-5-2:6
Relevance score	0.7586
Is stub	false

Excerpt:

for the non-3GPP access, these PDU Sessions are not released to enable the UE to move the PDU Sessions over the 3GPP access based on UE policies. The core network maintains the PDU Sessions but deactivates the N3 user plane connection for such PDU Sessions.

A.T0.technical.3 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.33.1

Field	Value
Source title	3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19
Authority	3rd Generation Partnership Project (3GPP)
Jurisdiction	International

Field	Value
Document type	Standard
Section	Clause 5.33.1
Section title	General
Page	<i>(empty)</i>
Date issued	2026-09
Effective	null
Effective status	<i>(empty)</i>
Amendment checked	false
Amendment note	<i>(empty)</i>
Url	https://www.3gpp.org/ftp/Specs/archive/23_series/23.501/23501-j90.zip
Provenance note	Official archive 23501-j90.zip from 3gpp.org (latest Release 19 version on 2026-10-05; the DOCX does not fix a version). Body paragraphs only; tables and figures not ingested. Reference point only, not Indian law.
Chunk id	3gpp_ts_23501:clause-5-33-1:1
Relevance score	0.7002
Is stub	false

Excerpt:

The following features described in 5.33 may be used to enhance 5GS to support Ultra Reliable Low Latency Communication (URLLC): - Redundant transmission for high reliability communication. In this Release, URLLC applies to 3GPP access only. When a PDU Session is to serve URLLC QoS Flow, the UE and SMF should establish the PDU Session as always-on PDU Session as described in clause 5.6.13. NOTE 1: How the UE knows whether a PDU Session is to serve a URLLC QoS Flow when triggering PDU Session establishment is up to UE implementation. NOTE 2: No additional functionality is specified for URLLC in order to support Home Routed roaming scenario in this Release.

A.T0.technical.4 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.45.2

Field	Value
Source title	3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19
Authority	3rd Generation Partnership Project (3GPP)
Jurisdiction	International
Document type	Standard
Section	Clause 5.45.2
Section title	Packet delay monitoring

Field	Value
Page	(empty)
Date issued	2026-09
Effective	null
Effective status	(empty)
Amendment checked	false
Amendment note	(empty)
Url	https://www.3gpp.org/ftp/Specs/archive/23_series/23.501/23501-j90.zip
Provenance note	Official archive 23501-j90.zip from 3gpp.org (latest Release 19 version on 2026-10-05; the DOCX does not fix a version). Body paragraphs only; tables and figures not ingested. Reference point only, not Indian law.
Chunk id	3gpp_ts_23501:clause-5-45-2:1
Relevance score	0.7214
Is stub	false

Excerpt:

QoS Monitoring for packet delay allows for the measurement of UL packet delay, DL packet delay or round trip packet delay between UE and PSA UPF. The details of the QoS Monitoring for packet delay are described in clause 5.33.3. The PCF may calculate Packet Delay Variation (clause 5.37.7) and the round trip packet delay when UL and DL are on different QoS flows (clause 5.37.4) based on packet delay monitoring results of QoS flows.

A.T0.technical.5 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.7.5.1

Field	Value
Source title	3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19
Authority	3rd Generation Partnership Project (3GPP)
Jurisdiction	International
Document type	Standard
Section	Clause 5.7.5.1
Section title	General
Page	(empty)
Date issued	2026-09
Effective	null
Effective status	(empty)
Amendment checked	false

Field	Value
Amendment note	(empty)
Url	https://www.3gpp.org/ftp/Specs/archive/23_series/23.501/23501-j90.zip
Provenance note	Official archive 23501-j90.zip from 3gpp.org (latest Release 19 version on 2026-10-05; the DOCX does not fix a version). Body paragraphs only; tables and figures not ingested. Reference point only, not Indian law.
Chunk id	3gpp_ts_23501:clause-5-7-5-1:2
Relevance score	0.7347
Is stub	false

Excerpt:

and PDU Sessions transferred from EPS without N26 interface, the UE indicates Reflective QoS support using the PDU Session Establishment procedure. After the first inter-system change from EPS to 5GS for PDU Sessions established in EPS and transferred from EPS with N26 interface, the UE indicates Reflective QoS support using the PDU Session Modification procedure as described in clause 5.17.2.2.2. The UE as well as the network shall apply the information whether or not the UE indicated support of Reflective QoS throughout the lifetime of the PDU Session. NOTE: The logic driving a supporting UE under exceptional circumstances to not indicate support of Reflective QoS for a PDU Session is implementation dependent. Under exceptional circumstances, which are UE implementation dependent, the UE may decide to revoke previously indicated support for Reflective QoS using the PDU Session Modification procedure. In such a case, the UE shall delete all derived QoS rules for this PDU Session and the network shall stop any user plane enforcement actions related to Reflective QoS for this PDU Session.

A.T0.technical.6 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.7.3.4

Field	Value
Source title	3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19
Authority	3rd Generation Partnership Project (3GPP)
Jurisdiction	International
Document type	Standard
Section	Clause 5.7.3.4
Section title	Packet Delay Budget
Page	(empty)
Date issued	2026-09
Effective	null
Effective status	(empty)
Amendment checked	false
Amendment note	(empty)

Field	Value
Url	https://www.3gpp.org/ftp/Specs/archive/23_series/23.501/23501-j90.zip
Provenance note	Official archive 23501-j90.zip from 3gpp.org (latest Release 19 version on 2026-10-05; the DOCX does not fix a version). Body paragraphs only; tables and figures not ingested. Reference point only, not Indian law.
Chunk id	3gpp_ts_23501:clause-5-7-3-4:5
Relevance score	0.6901
Is stub	false

Excerpt:

packet drops. Packets surviving congestion related packet dropping may still be subject to non-congestion related packet losses (see PER below). Services using Non-GBR QoS Flows should be prepared to experience congestion-related packet drops and delays. In uncongested scenarios, 98 percent of the packets should not experience a delay exceeding the 5QI's PDB. The PDB for Non-GBR and GBR resource types denotes a "soft upper bound" in the sense that an "expired" packet, e.g. a link layer SDU that has exceeded the PDB, does not need to be discarded and is not added to the PER. However, for a Delay-critical GBR resource type, packets delayed more than the PDB are added to the PER and can be discarded or delivered depending on local decision.

A.T0.technical.7 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 3.1

Field	Value
Source title	3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19
Authority	3rd Generation Partnership Project (3GPP)
Jurisdiction	International
Document type	Standard
Section	Clause 3.1
Section title	Definitions
Page	(empty)
Date issued	2026-09
Effective	null
Effective status	(empty)
Amendment checked	false
Amendment note	(empty)
Url	https://www.3gpp.org/ftp/Specs/archive/23_series/23.501/23501-j90.zip
Provenance note	Official archive 23501-j90.zip from 3gpp.org (latest Release 19 version on 2026-10-05; the DOCX does not fix a version). Body paragraphs only; tables and figures not ingested. Reference point only, not Indian law.

Field	Value
Chunk id	3gpp_ts_23501:clause-3-1:14
Relevance score	0.7158
Is stub	false

Excerpt:

Function: A 3GPP adopted or 3GPP defined processing function in a network, which has defined functional behaviour and 3GPP defined interfaces. NOTE 2: A network function can be implemented either as a network element on a dedicated hardware, as a software instance running on a dedicated hardware, or as a virtualised function instantiated on an appropriate platform, e.g. on a cloud infrastructure. Network Instance: Information identifying a domain. Used by the UPF for traffic detection and routing. Network Slice: A logical network that provides specific network capabilities and network characteristics. Network Slice Area of Service (NS-AoS): The area where a network slice is available i.e. the UE can access and get service of a particular network slice as more than zero resources are allocated to the network slice in the NG-RAN cells. This area may be, depending on the specific network slice, the whole PLMN, one or more TAs, or one or more cells when the NS-AoS does not match deployed TAs as defined in clause 5.15.18. Network Slice instance: A set of Network Function instances and the required resources (e.g.

A.T0.technical.8 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.17.3

Field	Value
Source title	3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19
Authority	3rd Generation Partnership Project (3GPP)
Jurisdiction	International
Document type	Standard
Section	Clause 5.17.3
Section title	Interworking with EPC in presence of Non-3GPP PDU Sessions
Page	(empty)
Date issued	2026-09
Effective	null
Effective status	(empty)
Amendment checked	false
Amendment note	(empty)
Url	https://www.3gpp.org/ftp/Specs/archive/23_series/23.501/23501-j90.zip
Provenance note	Official archive 23501-j90.zip from 3gpp.org (latest Release 19 version on 2026-10-05; the DOCX does not fix a version). Body paragraphs only; tables and figures not ingested. Reference point only, not Indian law.
Chunk id	3gpp_ts_23501:clause-5-17-3:1

Field	Value
Relevance score	0.7346
Is stub	false

Excerpt:

When a UE is simultaneously connected to the 5GC over a 3GPP access and a non-3GPP access, it may have PDU Sessions associated with 3GPP access and PDU Sessions associated with non-3GPP access. When inter-system handover from 5GS to EPS is performed for PDU Sessions associated with 3GPP access, the PDU Sessions associated with non-3GPP access are kept anchored by the network in 5GC and the UE may either: - keep PDU Sessions associated with non-3GPP access in 5GS (5GC+N3IWF or TNGF) (i.e. the UE is then registered both in EPS and, for non-3GPP access, in 5GS); or - locally or explicitly release PDU Sessions associated with non-3GPP access; or - once in EPS, transfer PDU Sessions associated with non-3GPP access to E-UTRAN by triggering PDN connection establishment with Request Type "Handover", as specified in TS 23.401 [26].

A.T0.technical.9 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 4.2.12

Field	Value
Source title	3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19
Authority	3rd Generation Partnership Project (3GPP)
Jurisdiction	International
Document type	Standard
Section	Clause 4.2.12
Section title	Architecture for Proximity based Services (ProSe) in 5GS
Page	(empty)
Date issued	2026-09
Effective	null
Effective status	(empty)
Amendment checked	false
Amendment note	(empty)
Url	https://www.3gpp.org/ftp/Specs/archive/23_series/23.501/23501-j90.zip
Provenance note	Official archive 23501-j90.zip from 3gpp.org (latest Release 19 version on 2026-10-05; the DOCX does not fix a version). Body paragraphs only; tables and figures not ingested. Reference point only, not Indian law.
Chunk id	3gpp_ts_23501:clause-4-2-12:1
Relevance score	0.682
Is stub	false

Excerpt:

The architecture for Proximity based Services (ProSe) in the 5G System is defined in TS 23.304 [128].

A.T0.technical.10 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.33.3.3

Field	Value
Source title	3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19
Authority	3rd Generation Partnership Project (3GPP)
Jurisdiction	International
Document type	Standard
Section	Clause 5.33.3.3
Section title	GTP-U Path Measurement
Page	<i>(empty)</i>
Date issued	2026-09
Effective	null
Effective status	<i>(empty)</i>
Amendment checked	false
Amendment note	<i>(empty)</i>
Url	https://www.3gpp.org/ftp/Specs/archive/23_series/23.501/23501-j90.zip
Provenance note	Official archive 23501-j90.zip from 3gpp.org (latest Release 19 version on 2026-10-05; the DOCX does not fix a version). Body paragraphs only; tables and figures not ingested. Reference point only, not Indian law.
Chunk id	3gpp_ts_23501:clause-5-33-3-3:4
Relevance score	0.7141
Is stub	false

Excerpt:

UL/DL packet delay by combining the received measurements of RAN part with the measurements of N3/N9 interface (N9 is applicable when I-UPF exists). - The UPF reports the QoS Monitoring results as described in clause 5.8.2.18. QoS Monitoring can also be used to measure the packet delay for transport paths to influence the mapping of QoS Flows to appropriate network instances, DSCP values as follows: - SMF activates QoS monitoring for the GTP-U path(s) between all UPF(s) and all (R)AN nodes based on locally configured policies. - UPF does measurement of network hop delay per transport resources that it will use towards a peer network node identified by an IP destination address (the hop between these two nodes) and port. The network hop measured delay is computed by sending an Echo Request over such transport resource (Ti) and measuring RTT/2 when Echo Response is received. - UPF maps

{network instance, DSCP} into Transport Resource and measures delay per IP destination address and port. Thus, for each IP destination address, the measured delay per (network instance, DSCP) entry is determined.

A.T0.technical.11 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.6.4.3

Field	Value
Source title	3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19
Authority	3rd Generation Partnership Project (3GPP)
Jurisdiction	International
Document type	Standard
Section	Clause 5.6.4.3
Section title	Usage of IPv6 multi-homing for a PDU Session
Page	<i>(empty)</i>
Date issued	2026-09
Effective	null
Effective status	<i>(empty)</i>
Amendment checked	false
Amendment note	<i>(empty)</i>
Url	https://www.3gpp.org/ftp/Specs/archive/23_series/23.501/23501-j90.zip
Provenance note	Official archive 23501-j90.zip from 3gpp.org (latest Release 19 version on 2026-10-05; the DOCX does not fix a version). Body paragraphs only; tables and figures not ingested. Reference point only, not Indian law.
Chunk id	3gpp_ts_23501:clause-5-6-4-3:1
Relevance score	0.7141
Is stub	false

Excerpt:

A PDU Session may be associated with multiple IPv6 prefixes. This is referred to as multi-homed PDU Session. The multi-homed PDU Session provides access to the Data Network via more than one PDU Session Anchor. The different user plane paths leading to the different PDU Session Anchors branch out at a "common" UPF referred to as a UPF supporting "Branching Point" functionality. The Branching Point provides forwarding of UL traffic towards the different PDU Session Anchors and merge of DL traffic to the UE i.e. merging the traffic from the different PDU Session Anchors on the link towards the UE. The UPF supporting a Branching Point functionality may also be controlled by the SMF to support traffic measurement for charging, traffic replication for LI and bit rate enforcement (Session-AMBR per PDU Session). The insertion and removal of a UPF supporting Branching Point is decided by the SMF and controlled by the SMF using generic N4 and UPF capabilities.

A.T0.technical.12 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.27.1.12

Field	Value
Source title	3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19
Authority	3rd Generation Partnership Project (3GPP)
Jurisdiction	International
Document type	Standard
Section	Clause 5.27.1.12
Section title	Support for network timing synchronization status monitoring
Page	(empty)
Date issued	2026-09
Effective	null
Effective status	(empty)
Amendment checked	false
Amendment note	(empty)
Url	https://www.3gpp.org/ftp/Specs/archive/23_series/23.501/23501-j90.zip
Provenance note	Official archive 23501-j90.zip from 3gpp.org (latest Release 19 version on 2026-10-05; the DOCX does not fix a version). Body paragraphs only; tables and figures not ingested. Reference point only, not Indian law.
Chunk id	3gpp_ts_23501:clause-5-27-1-12:1
Relevance score	0.6762
Is stub	false

Excerpt:

While the time synchronization service is offered by the 5GS, based on 5G access stratum-based time distribution or (g)PTP-based time distribution, the network timing synchronization status of the nodes involved in the operation (e.g. gNBs and/or UPF/NW-TTs) may change. gNBs and UPF/NW-TT can detect timing synchronization degradation or improvement locally. The support for network timing synchronization status monitoring enables the 5GS to modify time synchronization service for a UE or a group of UEs depending on the current synchronization status and notify service updates. There may be three consumers of this information: - TSCTSF may receive node-level information about timing synchronization status from gNB and/or UPF/NW-TT directly from OAM or alternatively, if supported by a node, using control plane signalling at node level. Node level signalling uses UMIC for UPF/NW-TT case and an AMF service to report N2 node level information for the gNB case.

A.T0.technical.13 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.15.1

Field	Value
Source title	3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19
Authority	3rd Generation Partnership Project (3GPP)
Jurisdiction	International
Document type	Standard
Section	Clause 5.15.1
Section title	General
Page	<i>(empty)</i>
Date issued	2026-09
Effective	null
Effective status	<i>(empty)</i>
Amendment checked	false
Amendment note	<i>(empty)</i>
Url	https://www.3gpp.org/ftp/Specs/archive/23_series/23.501/23501-j90.zip
Provenance note	Official archive 23501-j90.zip from 3gpp.org (latest Release 19 version on 2026-10-05; the DOCX does not fix a version). Body paragraphs only; tables and figures not ingested. Reference point only, not Indian law.
Chunk id	3gpp_ts_23501:clause-5-15-1:5
Relevance score	0.7107
Is stub	false

Excerpt:

as described in clause 5.15.13. The selection of N3IWF/TNGF supporting a set of slice(s) is described in clause 6.3.6 and clause 6.3.12 respectively. The support of Network Slice usage control is described in clause 5.15.15. Support of Optimized handling of temporarily available network slices is described in clause 5.15.16. It also covers aspects related to graceful release of network slices connectivity during slice decommissioning. The Partial Network Slice support in a Registration Area is described in clause 5.15.17. Support for Network Slices with Network Slice Area of Service not matching deployed Tracking Areas is described in clause 5.15.18. Support of Network Slice Replacement is described in clause 5.15.19.

A.T0.technical.14 3GPP TS 33.501 V19.7.0 (2026-06) Security architecture and procedures for 5G system, Release 19, Clause M.4

Field	Value
Source title	3GPP TS 33.501 V19.7.0 (2026-06) Security architecture and procedures for 5G system, Release 19
Authority	3rd Generation Partnership Project (3GPP)
Jurisdiction	International
Document type	Standard

Field	Value
Section	Clause M.4
Section title	Protection of management traffic between IAB-node and OAM
Page	<i>(empty)</i>
Date issued	2026-06
Effective	null
Effective status	<i>(empty)</i>
Amendment checked	false
Amendment note	<i>(empty)</i>
Url	https://www.3gpp.org/ftp/Specs/archive/33_series/33.501/33501-j70.zip
Provenance note	Official archive 33501-j70.zip from 3gpp.org (latest Release 19 version on 2026-10-05; the DOCX does not fix a version). Body paragraphs only; tables and figures not ingested. Reference point only, not Indian law.
Chunk id	3gpp_ts_33501:clause-m-4:1
Relevance score	0.7135
Is stub	false

Excerpt:

If management traffic uses the user plane via PDU session, it shall be protected using the user plane security mechanism as specified in clause 6.6.

Appendix B. Claims and citations in full

Stage T0: Technical agent

Claim T0.technical.1: ● UNSUPPORTED

The incident involves Network Slice (performance layer), User Plane Function (UPF), Private 5G Network (dedicated slice).

- **Verifier rationale:** Checked against any retrieved Canonical KB passage: the authoritative text does not support the claim (the claim's content is not found in the passage).
- **Cross-domain flag:** false
- **Cross-domain note:** *(empty)*

Citations given by the agent:

None.

Passages the Verifier found supporting:

None.

Passages the Verifier checked that do not support the claim:

Source	Section	Page	Chunk ID	Other
3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19	Clause 3.1	<i>(empty)</i>	3gpp_ts_23501:clause-3-1:14	<i>(empty)</i>
3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19	Clause 5.20	<i>(empty)</i>	3gpp_ts_23501:clause-5-20:19	<i>(empty)</i>
3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19	Clause 5.15.1	<i>(empty)</i>	3gpp_ts_23501:clause-5-15-1:1	<i>(empty)</i>

Claim T0.technical.2: ● UNSUPPORTED

Incident classification: Network performance degradation / potential service-quality failure.

- **Verifier rationale:** Checked against any retrieved Canonical KB passage: the authoritative text does not support the claim (the claim's content is not found in the passage).
- **Cross-domain flag:** false
- **Cross-domain note:** *(empty)*

Citations given by the agent:

None.

Passages the Verifier found supporting:

None.

Passages the Verifier checked that do not support the claim:

Source	Section	Page	Chunk ID	Other
ENISA 5G Cybersecurity Standards — Analysis of standardisation requirements in support of cybersecurity policy	Section 4.3	28-30	enisa_5g_cybersecurity_standards_2022:section-4-3:4	(empty)
ENISA 5G Cybersecurity Standards — Analysis of standardisation requirements in support of cybersecurity policy	Section 4.2	22-28	enisa_5g_cybersecurity_standards_2022:section-4-2:7	(empty)
ENISA 5G Cybersecurity Standards — Analysis of standardisation requirements in support of cybersecurity policy	Section 4.5	30-32	enisa_5g_cybersecurity_standards_2022:section-4-5:1	(empty)

Claim T0.technical.3: 🟡 INCOMPLETE

[REFERENCE ONLY — not Indian law] 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.37.7.1 states: "The 5GS Packet Delay Variation is the variation of packet delay measured between UE and PSA UPF. The AF may send the requirement for Packet Delay Variation monitoring to 5GS together with the requirement for packet delay measurement, as described in clause 6.1.3.26 of TS 23.503 [45]. Upon AF request for Packet Delay Variation monitoring together with packet delay monitoring, the PCF triggers the ..."

- **Verifier rationale:** Supported by the authoritative text (3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.37.7.1), but in-force status not verified and amendments not checked for this source.
- **Cross-domain flag:** false
- **Cross-domain note:** (empty)

Citations given by the agent:

Source	Section	Page	Chunk ID	Other
3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19	Clause 5.37.7.1	(empty)	3gpp_ts_23501:clause-5-37-7-1:1	(empty)

Passages the Verifier found supporting:

Source	Section	Page	Chunk ID	Other
3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19	Clause 5.37.7.1	(empty)	3gpp_ts_23501:clause-5-37-7-1:1	(empty)

Passages the Verifier checked that do not support the claim:

None.

Claim T0.technical.4: 🟡 INCOMPLETE

[REFERENCE ONLY — not Indian law] 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.5.2 states: "for the non-3GPP access, these PDU Sessions are not released to enable the UE to move the PDU Sessions over the 3GPP access based on UE policies. The core network maintains the PDU Sessions but deactivates the N3 user plane connection for such PDU Sessions."

- **Verifier rationale:** Supported by the authoritative text (3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.5.2; 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.5.2), but in-force status not verified and amendments not checked for this source.
- **Cross-domain flag:** false
- **Cross-domain note:** (*empty*)

Citations given by the agent:

Source	Section	Page	Chunk ID	Other
3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19	Clause 5.5.2	(<i>empty</i>)	3gpp_ts_23501:clause-5-5-2:6	(<i>empty</i>)

Passages the Verifier found supporting:

Source	Section	Page	Chunk ID	Other
3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19	Clause 5.5.2	(<i>empty</i>)	3gpp_ts_23501:clause-5-5-2:6	(<i>empty</i>)
3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19	Clause 5.5.2	(<i>empty</i>)	3gpp_ts_23501:clause-5-5-2:5	(<i>empty</i>)
3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19	Clause 5.5.2	(<i>empty</i>)	3gpp_ts_23501:clause-5-5-2:3	(<i>empty</i>)
3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19	Clause 5.5.2	(<i>empty</i>)	3gpp_ts_23501:clause-5-5-2:4	(<i>empty</i>)

Passages the Verifier checked that do not support the claim:

None.

Claim T0.technical.5: ● INCOMPLETE

[REFERENCE ONLY — not Indian law] 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.33.1 states: "The following features described in 5.33 may be used to enhance 5GS to support Ultra Reliable Low Latency Communication (URLLC): - Redundant transmission for high reliability communication. In this Release, URLLC applies to 3GPP access only. When a PDU Session is to serve URLLC QoS Flow, the UE and SMF should establish the PDU Session as always-on PDU Session as described in clause 5.6.13. NOTE ..."

- **Verifier rationale:** Supported by the authoritative text (3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.33.1), but in-force status not verified and amendments not checked for this source.
- **Cross-domain flag:** false
- **Cross-domain note:** (*empty*)

Citations given by the agent:

Source	Section	Page	Chunk ID	Other
3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19	Clause 5.33.1	(<i>empty</i>)	3gpp_ts_23501:clause-5-33-1:1	(<i>empty</i>)

Passages the Verifier found supporting:

Source	Section	Page	Chunk ID	Other
3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19	Clause 5.33.1	(empty)	3gpp_ts_23501:clause-5-33-1:1	(empty)

Passages the Verifier checked that do not support the claim:

None.

Claim T0.technical.6: INCOMPLETE

[REFERENCE ONLY — not Indian law] 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.45.2 states: "QoS Monitoring for packet delay allows for the measurement of UL packet delay, DL packet delay or round trip packet delay between UE and PSA UPF. The details of the QoS Monitoring for packet delay are described in clause 5.33.3. The PCF may calculate Packet Delay Variation (clause 5.37.7) and the round trip packet delay when UL and DL are on different QoS flows (clause 5.37.4) based on packet ..."

- **Verifier rationale:** Supported by the authoritative text (3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.45.2), but in-force status not verified and amendments not checked for this source.
- **Cross-domain flag:** false
- **Cross-domain note:** (empty)

Citations given by the agent:

Source	Section	Page	Chunk ID	Other
3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19	Clause 5.45.2	(empty)	3gpp_ts_23501:clause-5-45-2:1	(empty)

Passages the Verifier found supporting:

Source	Section	Page	Chunk ID	Other
3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19	Clause 5.45.2	(empty)	3gpp_ts_23501:clause-5-45-2:1	(empty)

Passages the Verifier checked that do not support the claim:

None.

Appendix C. LLM prompts and responses

Stage T0: Technical agent

Field	Value
Provider	ollama
Model	qwen2.5:7b-instruct
Status	0 claim(s) accepted, 2 rejected; accepted claims go to the Verifier
Latency ms	63025.2
Prompt sha256	03c1770c08a22385804ccf211a24e8eda892b95b95cfb7cf0e24127655e6cabcb

Token usage: input_tokens 2433, output_tokens 316

Passages offered to the LLM:

- 3gpp_ts_23501:clause-5-37-7-1:1
- 3gpp_ts_23501:clause-5-5-2:6
- 3gpp_ts_23501:clause-5-33-1:1
- 3gpp_ts_23501:clause-5-45-2:1
- 3gpp_ts_23501:clause-5-7-5-1:2
- 3gpp_ts_23501:clause-5-7-3-4:5
- 3gpp_ts_23501:clause-3-1:14
- 3gpp_ts_23501:clause-5-17-3:1

System prompt:

You are the Technical Agent in a multi-agent policy and legal adviser for 5G network incidents in India.

Your mandate: Technical diagnosis of 5G radio, core, network slicing, and network-function components. Incident classification and impact assessment from a purely technical perspective.

Rules:

1. Use ONLY the passages provided below. Do not use outside knowledge of any law, rule, standard or institution.
2. Every claim must cite at least one passage id from the list, and should reuse the key words of the cited passage so a reviewer can check it against that text.
3. A passage marked [REFERENCE ONLY – not Indian law] is an international reference point: never present it as an obligation in India.
4. Stay within your mandate. Do not decide legal questions: say what the passages say and how they may bear on the incident, for qualified human review.
5. Say plainly what is uncertain and what information is missing.

Answer with one JSON object and nothing else:

```
{"reasoning_summary": "<2-4 sentences: how the passages bear on the facts you were given>",  
  "claims": [{"text": "<one claim, under 60 words>", "cites": ["<passage id>"]}],  
  "uncertainties": ["<...>"],  
  "missing_information": ["<...>"]}
```

At most 4 claims.

User prompt:

Information released to you at this stage:

- Intermittent latency and session drops in the private 5G slice.
- A private 5G network slice is experiencing intermittent latency spikes.

- Session drops are occurring on the affected slice.
- The degradation is intermittent rather than a complete outage.
- No additional context about the cause or the service supported by the slice is available at this stage.

Information only you received:

- NOC dashboard: one-way latency on the private slice peaks at 180 ms against a 20 ms target; about 3% of PDU sessions on the slice were released abnormally in the last hour.

Incident flags: {"cyber_event_suspected": false, "cii_flagged": false, "data_exposure_suspected": false}

Retrieved passages:

[3gpp_ts_23501:clause-5-37-7-1:1] 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.37.7.1 [REFERENCE ONLY – not Indian law]: The 5GS Packet Delay Variation is the variation of packet delay measured between UE and PSA UPF. The AF may send the requirement for Packet Delay Variation monitoring to 5GS together with the requirement for packet delay measurement, as described in clause 6.1.3.26 of TS 23.503 [45]. Upon AF request for Packet Delay Variation monitoring together with packet delay monitoring, the PCF triggers the QoS monitoring procedure and obtains the UL, DL or RT QoS Monitoring result from the SMF. After receiving the QoS Monitoring result, the PCF derives the 5GS Packet Delay Variation based on the QoS Monitoring result and then reports to the AF/NEF both packet delay measurements and Packet Delay Variation. The details of the QoS Monitoring and QoS Monitoring for packet delay are described in clauses 5.45 and 5.33.3.

NOTE: The derivation of 5GS Packet Delay Variation by PCF, based on QoS Monitoring,

[3gpp_ts_23501:clause-5-5-2:6] 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.5.2 [REFERENCE ONLY – not Indian law]: for the non-3GPP access, these PDU Sessions are not released to enable the UE to move the PDU Sessions over the 3GPP access based on UE policies. The core network maintains the PDU Sessions but deactivates the N3 user plane connection for such PDU Sessions.

[3gpp_ts_23501:clause-5-33-1:1] 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.33.1 [REFERENCE ONLY – not Indian law]: The following features described in 5.33 may be used to enhance 5GS to support Ultra Reliable Low Latency Communication (URLLC): - Redundant transmission for high reliability communication. In this Release, URLLC applies to 3GPP access only. When a PDU Session is to serve URLLC QoS Flow, the UE and SMF should establish the PDU Session as always-on PDU Session as described in clause 5.6.13. NOTE 1: How the UE knows whether a PDU Session is to serve a URLLC QoS Flow when triggering PDU Session establishment is up to UE implementation. NOTE 2: No additional functionality is specified for URLLC in order to support Home Routed roaming scenario in this Release.

[3gpp_ts_23501:clause-5-45-2:1] 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.45.2 [REFERENCE ONLY – not Indian law]: QoS Monitoring for packet delay allows for the measurement of UL packet delay, DL packet delay or round trip packet delay between UE and PSA UPF. The details of the QoS Monitoring for packet delay are described in clause 5.33.3. The PCF may calculate Packet Delay Variation (clause 5.37.7) and the round trip packet delay when UL and DL are on different QoS flows (clause 5.37.4) based on packet delay monitoring results of QoS flows.

[3gpp_ts_23501:clause-5-7-5-1:2] 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.7.5.1 [REFERENCE ONLY – not Indian law]: and PDU Sessions transferred from EPS without N26 interface, the UE indicates Reflective QoS support using the PDU Session Establishment procedure. After the first inter-system change from EPS to 5GS for PDU Sessions established in EPS and transferred from EPS with N26 interface, the UE indicates Reflective QoS support using the PDU Session Modification procedure as described in clause 5.17.2.2.2. The UE as well as the network shall apply the information whether or not the UE indicated support of Reflective QoS throughout the lifetime of the PDU Session. NOTE: The logic driving a supporting UE under exceptional circumstances to not indicate support of Reflective QoS for a PDU Session is implementation dependent. Under exceptional circumstances, which are UE implementation dependent, the UE may decide to revoke previously indicated support for Reflective QoS using the PDU Session Modification

[3gpp_ts_23501:clause-5-7-3-4:5] 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.7.3.4 [REFERENCE ONLY – not Indian law]: packet drops. Packets surviving congestion related packet dropping may still be subject to non-congestion related packet losses (see PER below). Services using Non-GBR QoS Flows should be prepared to experience congestion-related packet drops and delays. In uncongested scenarios, 98 percent of the packets should not experience a delay exceeding the 5QI's PDB. The PDB for Non-GBR and GBR resource types denotes a "soft upper bound" in the sense that an "expired" packet, e.g. a link layer SDU that has exceeded the

PDB, does not need to be discarded and is not added to the PER. However, for a Delay-critical GBR resource type, packets delayed more than the PDB are added to the PER and can be discarded or delivered depending on local decision.

[3gpp_ts_23501:clause-3-1:14] 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 3.1 [REFERENCE ONLY – not Indian law]: Function: A 3GPP adopted or 3GPP defined processing function in a network, which has defined functional behaviour and 3GPP defined interfaces. NOTE 2: A network function can be implemented either as a network element on a dedicated hardware, as a software instance running on a dedicated hardware, or as a virtualised function instantiated on an appropriate platform, e.g. on a cloud infrastructure. Network Instance: Information identifying a domain. Used by the UPF for traffic detection and routing. Network Slice: A logical network that provides specific network capabilities and network characteristics. Network Slice Area of Service (NS-AoS): The area where a network slice is available i.e. the UE can access and get service of a particular network slice as more than zero resources are allocated to the network slice in the NG-RAN cells. This area may be, depending on the specific network slice.

[3gpp_ts_23501:clause-5-17-3:1] 3GPP TS 23.501 V19.9.0 (2026-09) System architecture for the 5G System (5GS), Release 19, Clause 5.17.3 [REFERENCE ONLY – not Indian law]: When a UE is simultaneously connected to the 5GC over a 3GPP access and a non-3GPP access, it may have PDU Sessions associated with 3GPP access and PDU Sessions associated with non-3GPP access. When inter-system handover from 5GS to EPS is performed for PDU Sessions associated with 3GPP access, the PDU Sessions associated with non-3GPP access are kept anchored by the network in 5GC and the UE may either: - keep PDU Sessions associated with non-3GPP access in 5GS (5GC+N3IWF or TNGF) (i.e. the UE is then registered both in EPS and, for non-3GPP access, in 5GS); or - locally or explicitly release PDU Sessions associated with non-3GPP access; or - once in EPS, transfer PDU Sessions associated with non-3GPP access to E-UTRAN by triggering PDN connection establishment with Request Type "Handover", as specified in TS 23.401 [26].

Raw LLM response:

```
{
  "reasoning_summary": "The intermittent latency and session drops in the private 5G slice suggest issues with packet delay and potential congestion. The NOC dashboard indicates that the one-way latency is significantly higher than the target, and 3% of PDU sessions were abnormally released, which could be due to congestion or other issues.",
  "claims": [
    {
      "text": "The high one-way latency (180 ms) and packet drops indicate potential congestion issues in the 5G network, which could be affecting the private slice's performance.",
      "cites": ["[3gpp_ts_23501:clause-5-45-2:1]", "[3gpp_ts_23501:clause-5-7-3-4:5]"]
    },
    {
      "text": "The 3% abnormal release of PDU sessions could be due to congestion or other issues, as non-GBR QoS flows are expected to experience packet drops in uncongested scenarios.",
      "cites": ["[3gpp_ts_23501:clause-5-7-3-4:5]"]
    }
  ],
  "uncertainties": [
    "The exact cause of the latency spikes and session drops is uncertain without further diagnostic information."
  ],
  "missing_information": [
    "The specific service supported by the private 5G slice and the potential impact on the service are not provided."
  ]
}
```

Appendix D. Inputs visible to each agent

Stage T0: Technical agent

Field	Value
Chunk id	scenario2_T0
Information released	Intermittent latency and session drops in the private 5G slice.

New facts:

- A private 5G network slice is experiencing intermittent latency spikes.
- Session drops are occurring on the affected slice.
- The degradation is intermittent rather than a complete outage.
- No additional context about the cause or the service supported by the slice is available at this stage.

Information only this agent received:

- NOC dashboard: one-way latency on the private slice peaks at 180 ms against a 20 ms target; about 3% of PDU sessions on the slice were released abnormally in the last hour.

Prior chunk IDs:

None.

Incident state passed to the agent:

- **Chunks processed:** *(empty)*
 - **Cyber event suspected:** false
 - **Cii flagged:** false
 - **Data exposure suspected:** false
 - **Verified findings:** *(empty)*
-

Appendix E. Entry records and hash chain

Source file `scenario2_20261010T050639Z.jsonl`. Each hash is the SHA-256 of the entry without its hash field; each `prev_hash` is the previous entry's hash.

S e q	Entr y	Run ID	Sce nario	Recorded at	Pipeline timestamp	prev_hash	hash
1	run_start ed	scenario2_20261010T050639Z	sce nari o2	2026-10-10T05:06:39.915152+00:00	—	000000000000000000000000 000000000000000000000000 000000000000000000000000	c060cb26f5eaf83b10550e64a3b857a130b335fd7d75a2c1f7bf760aa0e8e22
2	stage T0	scenario2_20261010T050639Z	sce nari o2	2026-10-10T05:07:47.808011+00:00	2026-10-10T05:07:47.807010+00:00	c060cb26f5eaf83b10550e64a3b857a130b335fd7d75a2c1f7bf760aa0e8e22	1921bba8035b3e8970a51b69a126d08edb6218f2709de1a5d456cf71922554a2

Decision support only. Legal interpretation, regulatory decisions and institutional action remain with qualified people.